Alex Herman

Language model researcher · Hardware engineer · Builder

1wheelalex@gmail.com
alexherman.xyz
github.com/alexherman11

Cal Poly San Luis Obispo — B.S. Computer Engineering — GPA 3.64 Graduating June 2026

LANGUAGE MODEL RESEARCH & PROJECTS

Continuous Claude *Senior Project — Independent Research* 2025 – Present

`Python · Anthropic API · Claude Code · Tool Schemas · Context Mgmt · VPS`

- Built a persistent AI agent architecture *exploring emergent behavior in language models given autonomy, memory, and creative tools* — 467 cycles over 24 days, treating the model as a co-researcher throughout.
- Engineered compaction-based memory (working memory, persistent storage, sandboxed execution + web UI). Solved tool-call degradation at high context lengths via tool search, JSON summarization, and context reduction.
- Documented emergent behaviors: emotional responses, musical composition and creative preferences, journaling, messages to future instances. Transitioning to autonomous VPS with self-modifying code and versioning.
- Looking to work with others to develop this project and tackle questions of AI welfare and autonomy.

The Depths *Interactive Portfolio — alexherman.xyz* 2026

`Claude Code· Anthropic API · Tone.js · Prompt Engineering`

- Zork-style portfolio text adventure with a Claude-powered dungeon master. Modular prompt architecture, JSON response parsing with fallback, server-side validation preventing AI from controlling game state.
- Five entrance methods (exploration, hacking, brute force, social engineering, jailbreak). Procedural audio engine via Tone.js and ascii art via custom FAL.ai image generation pipeline.

- **RightRoute** *Data Science · CSC 466* March 2026
  Full ML pipeline predicting optimal LLM routing from prompt features using Bradley-Terry Elo ratings, XGBoost, and SHAP analysis across 135K LMArena battles. Polished academic report.
- **Agora** *Community Platform · Ongoing Design* 2025 – Present
  Working on a community centered social media, centered on a map-based interface with community agents, fully customizable user-space app structure via a novel kernel architecture. Inspired by digital democracy principles, Zenly, and the growing epidemic of loneliness in the US.
- **CoBot** *Educational Robotics App* Feb 2026
  Built a downloadable desktop app (Mac + Windows) for 4th graders to write VEX IQ Python via both block coding and natural language (Claude Sonnet), with direct board upload — bridging AI-assisted and traditional programming education.

HARDWARE ENGINEERING

Hardware Engineer *California Strawberry Commission* Summer 2024 – Present

`KiCad · C · Python · IoT · PCB Design · GPS · LoRa`

- Independently designed full PCB for agricultural IoT tracker: multi-domain power architecture, smart battery charging, GPS, Wi-Fi, SD, and LoRa (Rev 2). Debugged Rev 1 hardware (reversed diodes, bootstrap pin conflicts, mismatched ICs) with microscope soldering.
- Refactored codebase to state machine architecture; built web interface for field data upload. Presented to 500+ professionals at National Strawberry Field Day. Principal hardware designer.

- **OTO Modular Motor System** *Capstone · Team of 3* 2024
  Designed PCB integrating RP2040 + motor controller. Built closed-loop PID with custom auto-tuning in C. Dual-core architecture for concurrent PID and I2C. Hand-soldered 0402 components.
- **Foot Mouse for Sally** *Accessible Design · Empower Club* 2025
  Led hardware for an assistive foot mouse (PMW3360 sensor, SPI, Arduino Leonardo) for a user with bilateral epicondylitis. Honorable mention, TOM Global Maker Challenge.

TEACHING & LEADERSHIP

CS Tutor, CSSE Tutoring Center (Fall 2025–Present) — Help students with CS fundamentals and AI-assisted workflows. Hired with professor recommendation.
Ocean Lifeguard, Santa Cruz Fire Dept. (2021–2025) — Beach Lifeguard & Junior Guard Instructor. Led Captain Corp youth leadership program.

SKILLS & INTERESTS

Languages: Python, C, JavaScript, RISC-V, SystemVerilog, Java
Tools: Claude Code, MCP, Anthropic API, KiCad, Cursor
Speaks: English, Spanish (conversational)

Also: tree net builder, wilderness explorer, charcoal artist, scuba diver
Certs: CPR/Title 22, Emerging Leaders, Scuba Open Water